THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · EASY LEVEL UNIT TEST SET 1

Chapters 6–10: **We Distribute, Yet Things Multiply, Proportional Reasoning-1, Fractions in Disguise, The Baudhayana–Pythagoras Theorem, Proportional Reasoning-2** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. The value of 99×13 is: 1

(a) 1287 **(b)** 1297 **(c)** 1274 **(d)** 1300

2. The ratio 12 : 18 in simplest form is: 1

(a) 2 : 3 **(b)** 3 : 3 **(c)** 2 : 4 **(d)** 3 : 2

3. $\frac{1}{4}$ expressed as a percentage is: 1

(a) 25% **(b)** 28% **(c)** 50% **(d)** 22%

4. The legs of a right triangle are 3 cm and 4 cm. Its hypotenuse (in cm) is: 1

(a) 4 **(b)** 7 **(c)** 5 **(d)** 6

5. The ratio 12:18 in its simplest form is: 1

(a) 3:3 **(b)** 3:2 **(c)** 2:4 **(d)** 2:3

6. The value of $7 \times (10 + 3)$ is: 1

(a) 98 **(b)** 210 **(c)** 91 **(d)** 73

7. If 2 : 3 = 8 : x, then x is: 1

(a) 16 **(b)** 8 **(c)** 12 **(d)** 14

8. 20% written as a fraction in lowest terms is: 1

(a) $\frac{100}{20}$ **(b)** $\frac{25}{100}$ **(c)** 20 **(d)** $\frac{1}{5}$

9. The hypotenuse of a right triangle is 5 cm and one leg is 3 cm. The other leg (in cm) is: 1

(a) 4 **(b)** 3 **(c)** 2 **(d)** 5

10. If 2:3 = 8:x, then x is: 1

(a) 12 **(b)** 7 **(c)** 11 **(d)** 16

11. Expanding $3(x + 4)$ gives: 1

(a) $3x + 7$ **(b)** $3x + 12$ **(c)** $x + 12$ **(d)** $3x + 4$

12. Are 2 : 3 and 8 : 12 in proportion? 1

(a) Yes **(b)** Only when the first terms are equal

(c) No **(d)** Cannot be determined

13. 10% of 250 is: 1

(a) 50 **(b)** 25 **(c)** 28 **(d)** 22

14. Which of the following is a Pythagorean triplet? 1

(a) 3, 5, 5 **(b)** 4, 5, 5 **(c)** 3, 4, 5 **(d)** 3, 4, 6

15. ₹ 400 is divided between two friends in the ratio 3:5. The larger share (in ₹) is: 1

(a) 250 **(b)** 270 **(c)** 150 **(d)** 200

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. Find 99×16 using the distributive property. 2

17. Write the ratio of 40 cm to 2 m in simplest form. 2

18. Express $\frac{3}{8}$ as a percentage. 2

19. The legs of a right triangle are 3 cm and 4 cm. Find its hypotenuse. 2

20. If 6 pens cost ₹ 90, find the cost of 15 pens. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. Expand and simplify $2(x + 3) + 4(x + 5)$. 3

22. Divide ₹ 1200 between Ria and Sia in the ratio 3 : 5. 3

23. An article costing ₹ 200 is sold for ₹ 250. Find the profit percent. 3

24. A ladder 5 m long leans against a wall. Its foot is 3 m from the wall. How high on the wall does it reach? 3

OR

The scale of a map is 1 cm = 12 km. Two cities are 5 cm apart on the map. Find the actual distance. Also find the map distance of two towns that are actually 96 km apart. 3

25. Expand $(x + 2)(x + 3)$. 3

OR

Check whether 3 : 4 and 15 : 20 are in proportion. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. Expand $(x + 2)(x + 4)$ and find its value when $x = 3$. 5

OR

Two numbers are in the ratio 3 : 5 and their sum is 48. Find the numbers and their difference. 5

27. A family earns ₹ 20000 a month. It spends 30% on food and 25% on rent. Find the amount spent on each item and the amount left. 5

OR

A rectangular field is 3 m long and 4 m broad. Find the length of its diagonal. A man walks along the diagonal instead of the two sides; how much distance does he save? 5

END OF QUESTION PAPER — Total: 50 Marks